

# Development of Augmented Forward Collision Warning System for Head-Up Display\*

Changrak Yoon, Kyongho Kim, Hye Sun Park, Min Woo Park, and Soon Ki Jung

**Abstract**—In this paper, we present an augmented forward collision warning system for Head-Up Display (HUD). The convergence of HUD and Augmented Reality (AR) needs challenge and make innovative application in automobile industry. We focus on an advance of Forward Collision Warning System (FCWS) by the fusion of HUD and AR. The proposed system detects the frontal vehicles and pedestrians, assesses the imminent danger caused by the detected results, and alerts the augmented warnings to support safe driving practices.

## I. INTRODUCTION

HUD has been used in modern fighters to present information without requiring pilots to look away from their usual view. In the automobile industry, GM employed HUD first time in 1988 for Oldsmobile Cutlass Supreme. HUD gains popularity since it can increase situation awareness and reduce driving workload by 0.2 seconds than Head-Down Display (HDD) [1]. However, most commercial HUD products display simple information such as vehicle speed and direction to the destination, which are conventionally displayed in the instrument cluster. In recent years, interesting research has focused on object detection [2]-[4] and AR based HUD [5]-[8] to increase the driver safety and convenience. Unlike existing HUD systems, the in-vehicle AR-HUD system displays information registered to the driver's view [9].

Drivers should beware of vehicles, road hazard, lanes, pedestrians and traffic signs while controlling vehicle speed and directions. All these works increase physical and mental workload which is dangerous especially for elderly and people with slow recognition and responses. Therefore, an alarm that alerts the driver of a danger in the road can help to minimize driver workload and reduce vehicle accidents. Besides the navigational information, it should help driver's hazard event detection if the safety information such as crossing pedestrians are detected from front view cameras and their positions are highlighted in the HUD according to the driver's view point in the form of AR.

In this paper, we present the augmented FCWS. The proposed system detects the vehicles and pedestrians using

stereo cameras and displays early augmented warnings through an HUD (see-through display) to alert driver for taking a dodge or braking maneuver prior to collision. In the below, we describe the design and implementation issues with the real road test results. Then, future plans are described, followed by the conclusion.

## II. AUGMENTED FORWARD COLLISION WARNING SYSTEM

### A. Overview of Proposed System

In the proposed system, stereo cameras are mounted on top of the experimental vehicle to capture the frontal images. The captured stereo images are used to detect vehicles and pedestrians. Once the detection is done, the threat is assessed by Time-To-Collision (TTC) calculation, and the visual properties for augmented warnings are decided according to the TTC. A speedometer is used to calculate the TTC by presenting the driving speed, and the feasibility study helps to make references of visual properties such as shape, color and so on. Projective transformation is performed to take the augmented warnings according to the driver's view point obtained by an eye tracker. Geometric modeling between a camera and a HUD is carried out in a projective transformation. Finally, the augmented warnings are rendered onto an HUD. Fig.1 shows the overview of the proposed system. Fig. 2 shows the configuration of the proposed system.

### B. Vehicle and Pedestrian Detection

The proposed system detects vehicles and pedestrians using equi-height mosaicking image to improve the execution time of the existing Graphic Processing Unit (GPU) accelerated approach without decreasing the detection accuracy [10], [11].

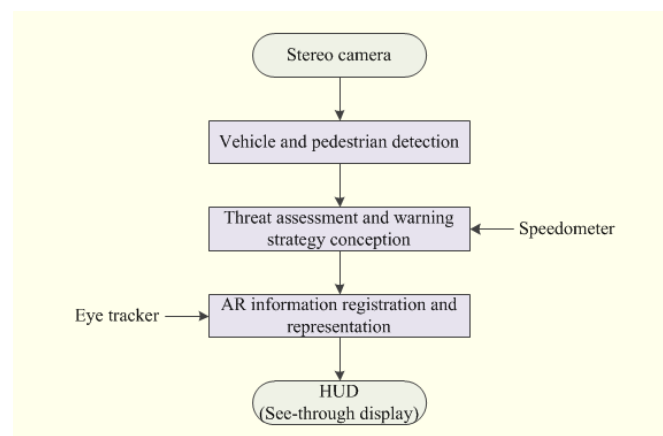


Figure 1. Overview of proposed system.

\*Research supported the Industrial Strategic Technology Development Program of MOTIE [10040927, Development of Driver-View based In-vehicle AR Display System Technology].

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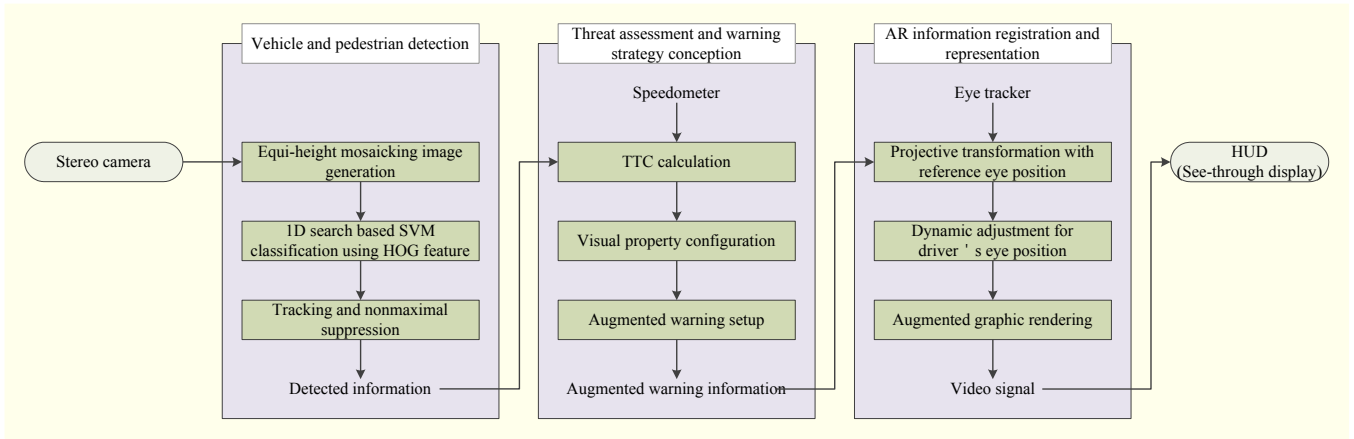


Figure 2. Configuration of proposed system.

The equi-height mosaicking image is an image strip cropped on the basis of the height of a normal vehicle (or pedestrian) in the image and concatenated in a panorama-like manner. The proposed system calibrates to compensate for the lens distortion and correct the image skew caused by the yawing of the camera. After the calibration is finished, road scene is analyzed to detect the vanishing point and horizon. The detected vanishing point and horizon are used to compute the height of the vehicle and pedestrian on the road plane in the perspective image. Hypothesis position sampling using bird's eye view is performed to extract the sampling position on the image at a uniform distance interval from the camera position. An equi-height image at each sampling position is generated and concatenated into an equi-height mosaicking image. The 1D search based the Support Vector Machine (SVM) classifier using the Histogram of Oriented Gradient (HOG) is trained and applied to detect vehicles and pedestrians. Tracking is performed by searching and matching only neighbor blocks of the detected position in the previous time. Finally, non-maximal suppression method is performed to group multiple overlapping candidates of vehicle and pedestrian.

### C. Threat assessment and warning strategy conception

The proposed system takes the distance values of the detected vehicles and pedestrians by performing a stereo vision. The driving velocity is obtained by using a speedometer. Dividing the distance values of the detected vehicles and pedestrians by the driving velocity, the proposed system acquires the TTC values. The calculated TTC values present the degree of safety. The augmented warnings for the detected vehicles and pedestrians depend on the calculated TTC values. The three threat level is decided by the calculated TTC values. According to the TCC values, the detected vehicles and pedestrians with high TCC values are used to represent the augmented warnings.

It is important to provide necessary information to the driver through an intuitive and helpful representation. To improve the driving safety and minimize the driving workload, the augmented warnings should be represented in such a way that it is more easily understood. It is therefore necessary to configure the characteristics of the augmented warnings, and

find a way to represent the augmented warnings according to these characteristics. In addition, the proposed system needs to be conducted into both the amount of information that can minimize a driver's cognitive load and the types of representation that can minimize the driver's visual load. The result of a feasibility study using driving simulator [12] is used to configure the visual properties for representation of the augmented warnings. Visual properties include the visual information of the threat level, shape, color, transparency, relative position and size, icon, text, and so on.

### D. AR Information Registration and representation

After the vehicles and pedestrians in imminent danger are distinguished by the threat assessment, the registration in HUD in the geometric perspective is performed to alerts the augmented warning according to the driver's view point. The vehicle coordinate system needs to be defined arbitrary. In the proposed system, the origin of the system is set in the back of the vehicle with the x-axis to the front. The y-axis is to the left direction and z-axis completes the right-handed coordinate system. Therefore, the system is constant regardless of vehicle's movement and attitude change. The other coordinate systems for the stereo cameras, eye tracker and HUD can be defined similarly. An approximation method with minimum complexity is used to overlay the augmented warnings on the real world. Geometric modeling between the 2D image of a camera and a 2D HUD is carried out in a projective transformation (2D homography) with a fixed driver's eye position. Because the projective transformation parameter is calculated in the fixed position and view, dynamic adjustment is performed to compensate the movements in the driver's eye position. Eye tracker is also used to recognize the driver's eye position in real time.

## III. EXPERIMENTS

In the proposed system, the stereo cameras are mounted on the top and the transparent display is installed in front of the passenger seat. The stereo cameras are 8 mm cameras with a resolution of  $640 \times 480$  at a speed of 30 fps and the transparent display is a 22-inch transparent Samsung LCD display with a transparency of 15% instead of an existing automotive HUD product. Before the road test, we create the projective

transformation parameters using the conjugate point acquisition. Current implementation let the user select detection target, e.g. vehicle or pedestrian, for real time processing. The visual properties can be modified to test various visual effects.

For training the SVM classifiers, we use a training data set that contained positive images of 1,067 vehicles, negative images of 38,253 non-vehicles, positive images of 7,248 pedestrians and negative images of 27,135 non-pedestrians. The accuracy is measured with an F1 measure. We compare the performance of the proposed system to the existing GPU-based OpenCV HOG detector and HOG-based SVM detector. Table 1 shows the experimental results.

Fig. 3 shows the images of the vehicles and pedestrians detected through the proposed system on the transparent display. The augmented warnings for the vehicles and pedestrians can be selectively highlighted in some situations for safe driving. Some of them should be helpful for people with slow recognition and responses like elderly.

The proposed system is displayed with delay when the detected object passes quickly. We can experience similar behaviors when the experimental vehicle turns quickly. Therefore, it needs to improve the performance and the speed, which responds to robustly about various and fast speed in various vehicle environments.

#### IV. CONCLUSION

We are in the first phase of research and development for the proposed system with the five year plans (2012-2016). Ending the second research year, an experimental application has been designed and tested. From the road tests, we can observe the potential of the augmented FCWS and derive further improvement issues. Future plan is derived as below.

The transparent display will be replaced by a projector HUD because the transparent display creates double vision (diplopia) causing frequent eye accommodation leading to eye strain. This replacement will require a conversion model between images and the curvy windshield for AR display such as the rational function model. The proposed system relies on a GPU accelerated computer which is not familiar with automotive industry. Therefore, it will be necessary to implement the proposed system in real time embedded systems using in automotive industry. The augmented warnings should be displayed based on the situation awareness. Since the augmented warnings often obstructs driver's field of vision of the road, the minimized and selected information will have to be displayed intelligently.

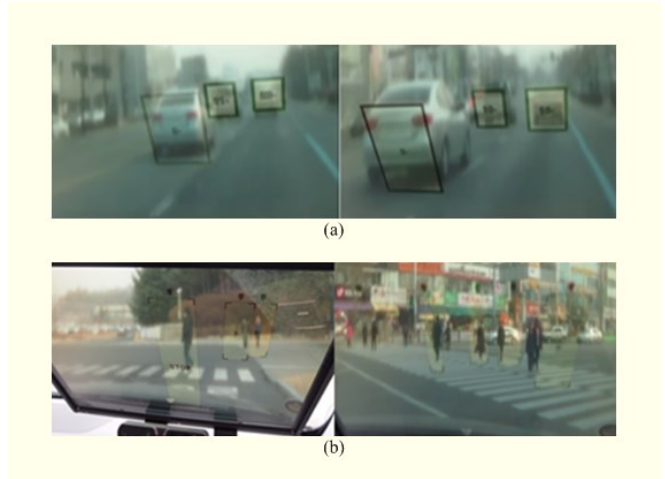


Figure 3. Augmented warning results for (a) vehicles and (b) pedestrians.

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TABLE I. VEHICLE AND PEDESTRIAN DETECTION RESULTS

| Object Type           | Experimental Results of Vehicle and Pedestrian Detection |                |           |         |            |
|-----------------------|----------------------------------------------------------|----------------|-----------|---------|------------|
|                       | Approach                                                 | Detection Time | Precision | Recall  | F1 measure |
| Vehicle (Highway)     | opencv-gpusvm                                            | 472 ms         | 82.14 %   | 80.20 % | 81.16 %    |
|                       | proposed method                                          | 47 ms          | 88.34 %   | 85.21 % | 86.75 %    |
| Vehicle (Downtown)    | opencv-gpusvm                                            | 497 ms         | 78.92 %   | 78.23 % | 78.57 %    |
|                       | proposed method                                          | 58 ms          | 86.25 %   | 83.67 % | 84.95 %    |
| Pedestrian (Downtown) | opencv-gpusvm                                            | 348 ms         | 77.48 %   | 72.26 % | 74.28 %    |
|                       | proposed method                                          | 83 ms          | 88.17 %   | 80.52 % | 84.17 %    |